

Abstract

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1 Introduction

1.1 Context and Motivation

1.2 Research Objectives

1.3 Main Contributions of the Thesis

1.4 Structure of the Document

2 Theoretical Background

The field of artificial intelligence has undergone deep transformations since its inception, evolving through distinct paradigms that reflect different philosophical and technical approaches to machine intelligence. This chapter provides the theoretical foundations necessary to understand the framework proposed in this thesis, exploring the evolution of learning paradigms in AI, some techniques of visual perception and the principles underlying symbolic reasoning and reinforcement learning.

2.1 The Evolution of Learning in Artificial Intelligence

2.1.1 Artificial Intelligence Paradigms

The history of artificial intelligence is characterized by the emergence and evolution of three major paradigms, each representing a distinct approach to providing machines the human-like reasoning abilities. The symbolic paradigm, dominant from the 1950s through the 1980s. This paradigm was grounded in the hypothesis that human intelligence could be replicated through the manipulation of symbols, according to logical rules, to perform reasoning tasks. The systems built on this paradigm operated on explicit knowledge representations, using formal logic and rule-based reasoning to solve problems. These systems are grounded in formal logic and manually designed rule sets. Classic examples include expert systems such as MYCIN and Deep Blue, as well as semantic networks and logic programming frameworks [1]. They are capable of performing reasoning tasks and have achieved notable success in constrained domains where knowledge can be explicitly encoded. However, the symbolic paradigm faced significant limitations due to its high reliance on manually encoded knowledge. Indeed, when faced with tasks requiring pattern recognition, learning from experience, the management of uncertainty or incomplete information, it suffered from brittle-

ness [1, 2]. These challenges gave rise to the connectionist paradigm, which gained prominence in the late 1980s and experienced a resurgence in the 2010s with the advent of deep learning [3]. Inspired by the structure and function of biological neural networks, connectionist systems learn representations directly from data through training processes that adjust connection weights between artificial neurons [3, 4]. This paradigm has achieved remarkable success in tasks such as image recognition, natural language processing and game playing, often surpassing human-level performance in specific domains [5]. Their capacity for robust and interpretable reasoning remains limited. Deep learning still struggles with structured reasoning, causal inference and factual consistency. Despite the impressive capabilities of neural networks, their operation and internal representation often lacks transparency, giving rise to what is commonly referred to as the "black box problem" [2], which limits interpretability, accountability and trust. Indeed, many of these approaches compromised interpretability and relied on strong assumptions. Moreover, they lacked the rich semantic expressiveness characteristic of symbolic systems. These limitations provide strong motivation for integrating neural and symbolic paradigms. In short, symbolic systems excel at reasoning but lack at perception, while neural networks excel at perception but struggle with reasoning [1]. This limitation has motivated the emergence of a hybrid paradigm, which seeks to combine the learning capabilities of neural networks with the interpretability and reasoning power of symbolic systems. Neuro-symbolic AI represents this third wave, aiming to bridge the gap between data-driven learning and knowledge-based reasoning as well as between logic-based inference and gradient-based learning. By integrating these paradigms, it aims to potentially offer systems that are both powerful and explainable [6]. Neuro-symbolic methods have demonstrated significant promise across a wide range of application areas, especially in scenarios that demand the combination of data-driven learning and structured symbolic reasoning [6]. This emerging paradigm reflects a synthesis of complementary approaches, bringing together the statistical strengths of deep learning, the formal foundations of symbolic logic and the adaptability of large-scale pretrained models.

Nonetheless, several important challenges remain. These include:

- Ensuring compatibility between the discrete symbolic structures used in symbolic approaches and the continuous vector representations typical of neural networks.
- Enabling the model to dynamically learn, adapt, modify, and refine logical rules.

- Achieving an optimal balance between system complexity and computational efficiency.
- Addressing the absence of unified architectural frameworks by developing a general architecture that systematically and reusably integrates neural and symbolic components.

Although several challenges persist, neuro-symbolic AI is widely regarded as a promising direction for developing systems capable of transparent, reliable and generalizable reasoning.

- 2.1.2 Symbolic and Hybrid Approaches in Artificial Intelligence
- 2.1.3 Reinforcement Learning: Learning Through Experience
- 2.2 Computer Vision Techniques
 - 2.2.1 Classical Techniques for Video Analysis
 - 2.2.2 Visual Perception in Dynamic Environments
 - 2.2.3 Deep Learning for Feature Extraction from Video Streams
- 2.3 Knowledge Representation and Reasoning
 - 2.3.1 Symbolic Representations, Semantic Networks, and Interpretable Models
- 2.4 Reinforcement Learning Foundations
 - 2.4.1 Reinforcement Learning Principles and Deep Q-Networks
- 2.5 Atari-like Environments as Testbeds
 - 2.5.1 Characteristics and Challenges of 2D Game Environments

3 State of the Art and Challenges

3.1 Challenges in Symbolic Interpretation

3.1.1 The Illusion of Understanding

The Lack of World Models

3.1.2 The Black Box Problem

Making Sense of AI: Toward Explainability

3.1.3 Learning Paradigms and Explainable AI

Learning Without Understanding

Explainable AI: Goals and Approaches

Post-hoc Explainability

Intrinsic Interpretability

Why XAI is Not Enough

3.2 Cognitive Foundations

3.2.1 Human World Modelling and the Origins of Structured Cognition

3.2.2 Core Knowledge Theory: Cognitive Principles and Psychological Foundations

3.2.3 Definition and Relevance for World Understanding

3.2.4 Cognitive Models and Symbolic Representation

3.3 Positioning of the Proposed Framework

4 Framework Design - Proposed Framework

4.1 Conceptual Foundations

4.1.1 Evolution of the Idea

4.2 Proposed Implementation

4.2.1 Segmentation

4.2.2 Symbolic System as Knowledge Construction Module

4.2.3 From Symbolic Rules to a Playable Environment

4.2.4 Generic Environment as a Transitional Representation

4.2.5 Deep Q-Network in a Domain-Agnostic Environment

5 Implementation

5.1 End-to-End Pipeline

5.2 General Architecture of the Framework

5.2.1 Design Choices

5.2.2 Selected Techniques and Algorithms

5.2.3 Modular Structure of the System

Video Acquisition and Preprocessing Module

Feature Extraction Module

Symbolic Mapping Module - Symbolic Reasoning Engine

Integration of Core Knowledge for Semantic Understanding

RL Agent

5.2.4 Optimizations and Efficiency Considerations

6 Experimental Evaluation

6.1 Datasets and Testing Scenarios

6.1.1 Evaluation Metrics: Accuracy, Interpretability, Scalability

6.2 Experimental Results

6.2.1 Comparison with Existing Methods

6.2.2 Critical Discussion of the Results

7 Applications and Future Perspectives – Conclusion

7.1 Future Directions

7.1.1 Potential Applications of the Framework

7.1.2 Possible Extensions of the Work

7.1.3 Open Challenges and Future Research Directions

7.2 Summary of Contributions, Impact of the Research, and Final Reflections

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A Additional Technical Details

B Code Snippets or Pseudocode of Relevant Modules

C Dataset Specifications or Experimental Settings